Composing Piecewise and Nonlinear Functions

Evaluating a composition, simplifying one into a single rule, choosing the right piece of a piecewise function, and solving composition equations

The functions used throughout this worksheet.

$$f(x) = x^2 - 3 \qquad g(x) = 2x + 1 \qquad h(x) = \sqrt{x - 4}$$

$$p(x) = \begin{cases} x^2, & x < 2 \\ 5 - x, & x \geq 2 \end{cases} \qquad q(x) = \begin{cases} -x, & x < 0 \\ x^2 + 1, & x \geq 0 \end{cases}$$

Directions. $(f \circ g)(x)$ means $f(g(x))$: the *inner* function acts first. With a piecewise outer function, choose the piece using the **inner function's output**, not the original input. Simplify exactly — keep radicals and fractions — and write the requested value or rule on the answer line.

1. Find $(f \circ g)(2)$.

Show the inner value first, then substitute it into the outer function.

Answer: ________________

2. Find a single simplified rule for $(f \circ g)(x) = f(g(x))$.

Expand fully and collect like terms.

Answer: ________________

3. Find $p(g(-2))$.

(a) Compute the inner value $g(-2)$. **(b)** State which piece of p that value selects, and why. **(c)** Give $p(g(-2))$.

Answer: ________________

4. Solve $(f \circ g)(x) = 13$ for x.

Use your simplified rule from problem 2, set it equal to 13, and solve the resulting quadratic exactly. List every real solution on the answer line.

Answer: ________________

5. Find $(g \circ f)(3)$.

Compare your answer with problem 1: the same two functions in the other order. Say in one line what that comparison shows.

Answer: ________________

6. Find a single simplified rule for $(g \circ f)(x) = g(f(x))$.

Then state how this rule differs from the rule you found in problem 2, and what that says about whether composition is commutative.

Answer: ____________________

7. Consider $(p \circ g)(x) = p(2x + 1)$.

(a) The rule for p changes when its input reaches 2. Solve $2x + 1 = 2$ to find the value of x where the composition switches pieces, and write that value on the answer line.

(b) Using that cut point, write $(p \circ g)(x)$ as a piecewise rule in x, with the correct inequality on each piece.

Answer: ____________________

8. Find $(q \circ f)(1)$.

(a) Compute $f(1)$. **(b)** Explain which piece of q that value selects — be careful, the test compares the *inner output* with 0, not x with 0. **(c)** Give the value of the composition.

Answer: ________________

9. Solve $(g \circ f)(x) = 45$ for x.

Use your rule from problem 6. List every real solution on the answer line, and state in one line why this equation has two solutions while a linear composition would have one.

Answer: ________________

10. Find a single simplified rule for $(f \circ h)(x) = f(h(x))$.

(a) Simplify the rule as far as possible and write it on the answer line. **(b)** The simplified rule would make sense for every real x, but the composition does not. State the domain of $(f \circ h)$ and explain what the simplification threw away.

Answer: ________________

11. Consider $(h \circ f)(x) = \sqrt{x^2 - 7}$.

(a) A square root needs a non-negative radicand. Solve $x^2 - 7 = 0$ and write both boundary values on the answer line.

(b) Use those boundaries to write the domain of $(h \circ f)$ in interval notation, and say why the domain comes in two pieces.

Answer: ________________

12. Solve $(q \circ g)(x) = 5$ for x, where q is piecewise.

(a) *First piece.* q uses $-u$ when its input $u = 2x + 1$ is negative. Solve $-(2x + 1) = 5$, then check whether your solution really does make $2x + 1$ negative.

(b) *Second piece.* q uses $u^2 + 1$ when $u = 2x + 1 \geq 0$. Solve $(2x + 1)^2 + 1 = 5$, then test each candidate the same way.

(c) Write the complete solution set on the answer line, and explain why one candidate had to be thrown out.

Answer: ____________________